
TOPICS AND SUBTOPICS · TRACKS A–E

Software Engineering Curriculum Outline

A compact map of the complete learning hierarchy: each track, major curriculum group, and numbered syllabus section—without individual leaf-level study concepts.

5

TRACKS

58

MAJOR GROUPS

591

NUMBERED SECTIONS

Generated September 4, 2026 from the live website taxonomy. The complete curriculum currently contains 4661 study nodes; those leaf topics are intentionally omitted here.

DSA & Problem Solving

Algorithms, patterns, and interview coding execution. 14 major groups · 14 sections.

A1 — Programming for Interviews

A2 — Arrays & Strings

A3 — Searching & Sorting

A4 — Linked Lists

A5 — Stack & Queue

A6 — Trees

A7 — Heaps

A8 — Graphs

A9 — Recursion & Backtracking

A10 — Greedy

A11 — Dynamic Programming

A12 — Tries & Advanced Structures

A13 — Bit Manipulation

A14 — Coding Interview Execution

Frontend Engineering

JavaScript language (beginner → interview), TypeScript, React, browser APIs. 6 major groups · 165 sections.

B1 — JavaScript

43 NUMBERED SUBTOPICS

- B1.1** JavaScript Foundations
- B1.2** Variables & Declarations
- B1.3** Types & Values
- B1.4** Type Conversion & Coercion
- B1.5** Operators & Expressions
- B1.6** Control Flow
- B1.7** Strings
- B1.8** Numbers & Math
- B1.9** Functions
- B1.10** Scope & Lexical Environments
- B1.11** Hoisting & Temporal Dead Zone
- B1.12** Closures
- B1.13** Objects
- B1.14** Object Property Model
- B1.15** this, call, apply, bind
- B1.16** Prototypes & Prototype Chain
- B1.17** Classes & OOP
- B1.18** Arrays
- B1.19** Maps, Sets & Weak Collections
- B1.20** Destructuring, Spread & Modern Syntax
- B1.21** Iterables, Iterators & Generators
- B1.22** Symbols
- B1.23** Execution Context
- B1.24** Call Stack
- B1.25** JavaScript Runtime & Engine
- B1.26** Event Loop
- B1.27** Timers & Scheduling
- B1.28** Callbacks
- B1.29** Promises
- B1.30** Async/Await
- B1.31** Async Cancellation & Coordination
- B1.32** Error Handling
- B1.33** Debugging
- B1.34** Modules
- B1.35** JSON & Serialization
- B1.36** Dates & Time
- B1.37** Regular Expressions
- B1.38** Functional JavaScript
- B1.39** Memory Management & Garbage Collection
- B1.40** Performance Patterns
- B1.41** Proxy & Reflect
- B1.42** Binary Data
- B1.43** JavaScript Implementation Exercises

B2 — TypeScript

35 NUMBERED SUBTOPICS

- B2.1** Introduction & Setup
- B2.2** Simple Types & Annotations
- B2.3** Type Inference
- B2.4** Special Types
- B2.5** Arrays
- B2.6** Tuples
- B2.7** Object Types
- B2.8** Enums & Literal Constants
- B2.9** Interfaces
- B2.10** Type Aliases
- B2.11** Unions & Intersections
- B2.12** Literal Types & Discriminated Unions
- B2.13** Type Narrowing & Type Guards
- B2.14** Functions
- B2.15** Casting & Assertions
- B2.16** Classes
- B2.17** Access Modifiers & Encapsulation
- B2.18** Abstract Classes & implements
- B2.19** Generics
- B2.20** Utility Types
- B2.21** keyof, typeof & Indexed Access
- B2.22** Structural Typing & Declaration Merging
- B2.23** Index Signatures
- B2.24** satisfies & as const
- B2.25** Conditional Types & infer
- B2.26** Mapped & Template Literal Types
- B2.27** Type-Safe API Design
- B2.28** React + TypeScript
- B2.29** Node.js + TypeScript
- B2.30** Modules
- B2.31** tsconfig & Compiler Options
- B2.32** Tooling & Ecosystem
- B2.33** Migration & Decorators
- B2.34** Advanced Patterns
- B2.35** TypeScript Implementation Exercises

B3 — Browser & Web Platform

37 NUMBERED SUBTOPICS

- B3.1** Browser Architecture
- B3.2** Window, Document & BOM
- B3.3** DOM Fundamentals
- B3.4** DOM Events
- B3.5** HTML Parsing & Page Loading
- B3.6** Critical Rendering Path
- B3.7** Layout, Reflow, Paint & Compositing
- B3.8** Browser Event Loop & Rendering
- B3.9** requestAnimationFrame
- B3.10** Browser HTTP
- B3.11** Fetch API
- B3.12** AbortController & Cancellation
- B3.13** Same-Origin Policy
- B3.14** CORS
- B3.15** Cookies
- B3.16** Web Storage
- B3.17** IndexedDB
- B3.18** Browser Caching
- B3.19** Content Security Policy
- B3.20** Browser Security Fundamentals
- B3.21** Web Workers
- B3.22** Service Workers
- B3.23** WebSockets
- B3.24** Server-Sent Events
- B3.25** Observer APIs
- B3.26** Blob & File APIs
- B3.27** Browser Performance
- B3.28** Resource Loading & Performance
- B3.29** Navigation & SPA Browser Behavior
- B3.30** iframe & Cross-Document Communication
- B3.31** URL & URL APIs
- B3.32** Forms & Browser Input
- B3.33** Clipboard, Drag & Drop, Selection
- B3.34** Permissions & Device APIs
- B3.35** Accessibility Tree & Focus
- B3.36** Browser Developer Tools
- B3.37** Browser Interview Scenarios

B4 — React

24 NUMBERED SUBTOPICS

- B4.1** React Foundations
- B4.2** Props, State & Data Flow
- B4.3** Rendering & Reconciliation
- B4.4** Events
- B4.5** Forms
- B4.6** Hooks Fundamentals
- B4.7** Context
- B4.8** Refs & Imperative React
- B4.9** Portals & Advanced Component Patterns
- B4.10** Error Handling
- B4.11** Suspense & Async UI
- B4.12** Performance Engineering
- B4.13** Concurrent React
- B4.14** Data Fetching & Server State
- B4.15** State Management
- B4.16** React Router
- B4.17** React 19 & Modern React
- B4.18** Server Rendering & Server Components
- B4.19** Next.js Integration
- B4.20** React Testing
- B4.21** Accessibility in React
- B4.22** Internationalization
- B4.23** Legacy React & Existing Codebases
- B4.24** React Architecture & Production Engineering

B5 — CSS / UI Engineering

12 NUMBERED SUBTOPICS

- B5.1** Cascade & Selectors
- B5.2** Box Model & Flow
- B5.3** Display & Visibility
- B5.4** Flexbox
- B5.5** Grid
- B5.6** Positioning & Stacking
- B5.7** Units, Color & Typography
- B5.8** Responsive Design
- B5.9** Theming & Custom Properties
- B5.10** Motion
- B5.11** CSS Architecture
- B5.12** HTML & CSS Accessibility

B6 — Frontend System Design

14 NUMBERED SUBTOPICS

- B6.1** Interview Method
- B6.2** Application Architecture
- B6.3** Component & Widget LLD
- B6.4** State Architecture
- B6.5** API Layer
- B6.6** Frontend Authentication
- B6.7** Lists, Search & Pagination
- B6.8** Realtime UI
- B6.9** Frontend Performance
- B6.10** Caching & Offline
- B6.11** Frontend Security Architecture
- B6.12** Observability
- B6.13** A11y, i18n & Design Systems
- B6.14** HLD Case Studies & Microfrontends

Java & Backend Engineering

Java, Spring Boot, PostgreSQL, Redis, Kafka, AWS. 16 major groups · 255 sections.

C1 — Core Java

16 NUMBERED SUBTOPICS

- C1.1** Syntax & Types
- C1.2** Operators & Control Flow
- C1.3** Arrays & Methods
- C1.4** Strings & Wrappers
- C1.5** OOP Foundations
- C1.6** Inheritance & Polymorphism
- C1.7** Records, Enums & Nested Types
- C1.8** Immutability & Object Contract
- C1.9** Exceptions
- C1.10** Generics & Annotations
- C1.11** Collections Framework
- C1.12** Lists
- C1.13** Sets, Maps & Queues
- C1.14** Functional Java
- C1.15** I/O & Serialization
- C1.16** Modern Java Language

C2 — JVM

12 NUMBERED SUBTOPICS

- C2.1** JDK, Runtime & JVM
- C2.2** Bytecode & Class Files
- C2.3** JVM Architecture
- C2.4** Class Loading & Initialization
- C2.5** Runtime Memory Areas
- C2.6** Object Allocation
- C2.7** Interpreter & JIT
- C2.8** Garbage Collection Foundations
- C2.9** Garbage Collectors
- C2.10** Runtime Failures & Leaks
- C2.11** Diagnostics & Profiling
- C2.12** JVM Tuning & Measurement

C3 — Java Concurrency

17 NUMBERED SUBTOPICS

- C3.1** Concurrency Foundations
- C3.2** Thread API & Lifecycle
- C3.3** Race Conditions & Safe State
- C3.4** Java Memory Model
- C3.5** Intrinsic Monitors
- C3.6** Explicit Locks & AQS
- C3.7** Atomics & Lock-Free Programming
- C3.8** Coordination Utilities
- C3.9** Concurrent Collections
- C3.10** Producer–Consumer & Backpressure
- C3.11** Executors & Thread Pools
- C3.12** Future & CompletableFuture
- C3.13** Fork/Join & Data Parallelism
- C3.14** Liveness & Failure Analysis
- C3.15** Virtual Threads
- C3.16** Structured Concurrency & Scoped Values
- C3.17** Concurrency Testing

C4 — Networking & Web

19 NUMBERED SUBTOPICS

- C4.1** Network Foundations & Models
- C4.2** Devices, Media & Wireless
- C4.3** Ethernet & Data Link Layer
- C4.4** IP Addressing & Subnetting
- C4.5** Routing, NAT & Network Control
- C4.6** Transport Layer & Sockets
- C4.7** TCP Reliability & Performance
- C4.8** DNS & Service Discovery
- C4.9** TLS & HTTPS
- C4.10** HTTP Semantics
- C4.11** HTTP Connections, Caching & Delivery
- C4.12** HTTP/2, HTTP/3 & QUIC
- C4.13** Cookies, Sessions & Browser Boundaries
- C4.14** Proxies, Load Balancers & CDN
- C4.15** API & Realtime Protocols
- C4.16** Network Security Fundamentals
- C4.17** Cloud & Container Networking
- C4.18** Performance & Troubleshooting
- C4.19** Java Networking APIs

C5 — Spring Core

14 NUMBERED SUBTOPICS

- C5.1** IoC & the Container
- C5.2** ApplicationContext
- C5.3** Dependency Injection
- C5.4** Beans, Names & Scopes
- C5.5** Bean Lifecycle
- C5.6** Component Scanning
- C5.7** Java Configuration
- C5.8** Profiles & Environment
- C5.9** Wiring Conflicts & Cycles
- C5.10** Events, Resources & i18n
- C5.11** Aspect-Oriented Programming
- C5.12** Spring Proxies
- C5.13** Container Internals
- C5.14** Testing Spring Core

C6 — Spring Boot

20 NUMBERED SUBTOPICS

- C6.1** Boot Foundations & Project Structure
- C6.2** SpringApplication & Startup
- C6.3** Auto-configuration & Starters
- C6.4** Externalized Configuration
- C6.5** Embedded Web Runtime
- C6.6** Spring MVC Request Lifecycle
- C6.7** REST Controllers & HTTP Semantics
- C6.8** DTOs, JSON & API Boundaries
- C6.9** Validation & Data Binding
- C6.10** Exception Handling & Error Contracts
- C6.11** Filters, Interceptors & Web Hooks
- C6.12** API Versioning & Evolution
- C6.13** Pagination, Sorting & Collection APIs
- C6.14** HTTP Clients & Service Integration
- C6.15** Task Execution & Scheduling
- C6.16** Logging & Diagnostics
- C6.17** Actuator & Application Availability
- C6.18** Spring Boot Testing
- C6.19** AOT, Native Images & Runtime Efficiency
- C6.20** Production Lifecycle & Packaging

C7 — SQL & PostgreSQL

20 NUMBERED SUBTOPICS

- C7.1** Relational & PostgreSQL Foundations
- C7.2** SQL Syntax, Types & Expressions
- C7.3** Databases, Schemas & Table DDL
- C7.4** Keys, Constraints & Generated Values
- C7.5** CRUD & Data Modification
- C7.6** Filtering, Ordering & Conditional Logic
- C7.7** Functions, Aggregation & Grouping
- C7.8** Joins & Set Operations
- C7.9** Subqueries & CTEs
- C7.10** Window Functions & Advanced Querying
- C7.11** Data Modeling & Normalization
- C7.12** Transactions, ACID & Isolation
- C7.13** MVCC, Locks & Deadlocks
- C7.14** Indexes & Access Methods
- C7.15** Query Planning & Performance
- C7.16** PostgreSQL Types & Document Features
- C7.17** Views, Functions, Procedures & Triggers
- C7.18** Connectivity, Security & Data Safety
- C7.19** Partitioning & Large Tables
- C7.20** Replication, Vacuum & Operations

C8 — JPA / Hibernate

16 NUMBERED SUBTOPICS

- C8.1** ORM Foundations & Technology Boundaries
- C8.2** Entities, Identity & Mapping Basics
- C8.3** Identifiers, Generated Values & Value Types
- C8.4** Relationships, Ownership & Cascades
- C8.5** Persistence Context, Lifecycle & Flush
- C8.6** Dirty Checking, Write-behind & SQL Generation
- C8.7** Fetching, Lazy Loading & N+1 Control
- C8.8** Repositories, Queries & Projections
- C8.9** Transactions, Locking & Concurrency
- C8.10** Schema, Validation & Migrations
- C8.11** Caching, Performance & Observability
- C8.12** Inheritance, Polymorphism & Advanced Mapping
- C8.13** Spring Data JPA Internals & Extension Points
- C8.14** Testing Persistence
- C8.15** API Boundaries, DTOs & Production Patterns
- C8.16** Interview Scenarios & Troubleshooting

C9 — Security

15 NUMBERED SUBTOPICS

- C9.1** Authentication vs Authorization
- C9.2** Sessions, Cookies & Browser Storage
- C9.3** JWT Structure, Claims & Validation
- C9.4** OAuth 2.0 Grants & Delegation
- C9.5** OIDC Identity vs OAuth Authorization
- C9.6** Token Management, Rotation & Revocation
- C9.7** Spring Security Filter Chain (Boot 3 / Security 6)
- C9.8** RBAC, ABAC & Method Security
- C9.9** XSS & Content-Security-Policy
- C9.10** CSRF Mechanics
- C9.11** SQL Injection & Unsafe Persistence
- C9.12** CORS Misconfigurations
- C9.13** Password Hashing
- C9.14** Secrets Management & Key Scanning
- C9.15** OWASP Architecture & Cryptographic Failures

C10 — Redis

16 NUMBERED SUBTOPICS

- C10.1** Redis Foundations & Why It Is Fast
- C10.2** Environment, Configuration & Clients
- C10.3** Keys, Commands & Core Operations
- C10.4** Data Structures
- C10.5** Caching Patterns & Cache-Aside
- C10.6** TTL & Expiration
- C10.7** Cache Invalidation
- C10.8** Distributed Caching
- C10.9** Sessions, Tokens & Feature State
- C10.10** Rate Limiting
- C10.11** Distributed Locks & Inventory
- C10.12** Pub/Sub, Queues & Real-Time Messaging
- C10.13** Transactions, Pipelining & Scripting
- C10.14** Persistence & Backup
- C10.15** Replication, Partitioning & Clustering
- C10.16** Security, Memory & Operations

C11 — Kafka & Event-Driven Systems

16 NUMBERED SUBTOPICS

- C11.1** Kafka Foundations & Messaging Models
- C11.2** Event-Driven Architecture
- C11.3** Cluster Architecture & Brokers
- C11.4** Topics & Log Anatomy
- C11.5** Partitions & Parallelism
- C11.6** Producers
- C11.7** Consumers
- C11.8** Consumer Groups & Rebalancing
- C11.9** Offsets & Retention Semantics
- C11.10** Ordering Guarantees
- C11.11** Delivery Semantics
- C11.12** Idempotency, Retries & Dead-Letter Queues
- C11.13** Schemas, Serialization & Evolution
- C11.14** Kafka Internals & Storage
- C11.15** Connect, Streams & Ecosystem
- C11.16** Outbox, Saga & Production Operations

C12 — Backend Reliability

14 NUMBERED SUBTOPICS

- C12.1** Reliability Foundations & Failure Modes
- C12.2** Timeouts & Deadlines
- C12.3** Retries & Backoff
- C12.4** Idempotency
- C12.5** Rate Limiting & Throttling
- C12.6** Circuit Breakers
- C12.7** Bulkheads & Isolation
- C12.8** Backpressure
- C12.9** Graceful Degradation & Fallbacks
- C12.10** Health Checks & Safe Rollouts
- C12.11** Eventual Consistency & Compensation
- C12.12** Failure Recovery
- C12.13** SLOs, Error Budgets & Reliability Signals
- C12.14** Chaos, Overload & Production Playbooks

C13 — Testing

14 NUMBERED SUBTOPICS

- C13.1** Testing Strategy & the Pyramid
- C13.2** Unit Testing
- C13.3** JUnit 5
- C13.4** Mockito & Test Doubles
- C13.5** Integration Testing
- C13.6** API & Web-Layer Testing
- C13.7** Database Testing
- C13.8** Testcontainers
- C13.9** Contract Testing
- C13.10** Spring Boot Test Slices & Context
- C13.11** Async, Messaging & Time
- C13.12** Load, Performance & Resilience Tests
- C13.13** Coverage, Mutation & Quality Gates
- C13.14** CI Strategy & Production Confidence

C14 — Docker / DevOps

16 NUMBERED SUBTOPICS

- C14.1** DevOps Foundations & Lifecycle
- C14.2** Version Control for Delivery
- C14.3** Linux Fundamentals for Containers
- C14.4** Containers & Docker Architecture
- C14.5** Docker Images & Registries
- C14.6** Dockerfile & Image Optimization
- C14.7** Docker Compose & Local Stacks
- C14.8** Docker Networking
- C14.9** Volumes & Container Storage
- C14.10** CI/CD Concepts & Deployment Automation
- C14.11** Pipeline Tools: GitHub Actions & Jenkins
- C14.12** Kubernetes Orchestration
- C14.13** AWS ECS & Managed Containers
- C14.14** Infrastructure as Code
- C14.15** Monitoring & Logging for Delivery
- C14.16** DevSecOps & Container Security

C15 — AWS / Cloud

16 NUMBERED SUBTOPICS

- C15.1** Cloud Foundations & Global Infrastructure
- C15.2** IAM & Account Security
- C15.3** Secrets Manager & Credential Hygiene
- C15.4** EC2 Compute
- C15.5** Lambda & Serverless
- C15.6** Containers: ECS, EKS & ECR
- C15.7** S3 Object Storage
- C15.8** EBS, EFS & Storage Comparison
- C15.9** VPC Networking & Security
- C15.10** Load Balancing & Auto Scaling
- C15.11** Route 53 & CloudFront
- C15.12** RDS & Aurora
- C15.13** DynamoDB & Managed Data Stores
- C15.14** SQS, SNS & Application Integration
- C15.15** CloudWatch, CloudTrail & Operations
- C15.16** Cost, IaC & Well-Architected

C16 — Observability

14 NUMBERED SUBTOPICS

- C16.1** Observability Foundations
- C16.2** Structured Logging
- C16.3** Metrics Fundamentals
- C16.4** Golden Signals & SLIs
- C16.5** Distributed Tracing
- C16.6** Correlation & Context Propagation
- C16.7** OpenTelemetry
- C16.8** Prometheus
- C16.9** Grafana, Dashboards & Visualization
- C16.10** Alerting & On-Call
- C16.11** Frontend, RUM & Full-Stack Telemetry
- C16.12** Cardinality, Cost & Sampling
- C16.13** Profiling & Continuous Diagnostics
- C16.14** Incident Debugging Playbooks

LLD, System Design & Distributed Systems

OOD, patterns, LLD, HLD, distributed systems. 10 major groups · 80 sections.

D1 — Object-Oriented Design

D2 — Design Patterns

D3 — LLD Interview Problems

D4 — Distributed Systems Foundations

12 NUMBERED SUBTOPICS

- D4.1 System Design Foundations, HLD & LLD Boundaries
- D4.2 Requirements, Constraints & Trade-offs
- D4.3 Scale from One Server to Millions
- D4.4 Latency, Throughput & Performance Budgets
- D4.5 Load Balancing, Proxies & Traffic Routing
- D4.6 Availability, Reliability & Failure Models
- D4.7 CAP, PACELC & Consistency Models
- D4.8 Replication, Leaders & Quorums
- D4.9 Partitioning, Sharding & Consistent Hashing
- D4.10 Distributed Time, Ordering & Consensus
- D4.11 Delivery Semantics & Idempotency
- D4.12 Distributed Transactions & Workflows

D5 — Data & Storage Architecture

10 NUMBERED SUBTOPICS

- D5.1 Data Modeling & Access-Pattern-First Design
- D5.2 SQL, NoSQL & Database Selection
- D5.3 Database Replication & Read Scaling
- D5.4 Database Partitioning & Sharding
- D5.5 Block, File & Object Storage
- D5.6 Caching Architecture & Invalidation
- D5.7 CDN & Edge Delivery
- D5.8 Queues, Pub/Sub & Async Work
- D5.9 Event Streams & Data Pipelines
- D5.10 Search, Indexes & Probabilistic Structures

D6 — API & Communication Design

6 NUMBERED SUBTOPICS

- D6.1 API Contracts & Protocol Selection
- D6.2 Resource Modeling, Errors & Compatibility
- D6.3 Pagination, Filtering & Large Results
- D6.4 Real-Time Communication
- D6.5 API Gateway, Security & Rate Limiting
- D6.6 Service Communication & Discovery

D7 — Scalability, Reliability & Architecture

8 NUMBERED SUBTOPICS

- D7.1 Architecture Styles & Service Boundaries
- D7.2 Caching Strategy at System Scale
- D7.3 Overload Protection & Backpressure
- D7.4 Timeouts, Retries & Circuit Breakers
- D7.5 High Availability, Failover & Disaster Recovery
- D7.6 SLIs, SLOs & Production Readiness
- D7.7 Multi-Region Architecture
- D7.8 Security, Privacy, Cost & Sustainability

D8 — Capacity Estimation

4 NUMBERED SUBTOPICS

- D8.1 Back-of-the-Envelope Estimation Method
- D8.2 Traffic, QPS & Concurrency Estimation
- D8.3 Storage, Bandwidth & Cache Estimation
- D8.4 Capacity Planning & Bottleneck Analysis

D9 — System Design Interview Method

7 NUMBERED SUBTOPICS

- D9.1** The 45–60 Minute Interview Framework
- D9.2** Requirements & Scope in Interviews
- D9.3** Estimation, APIs & Data Model in Interviews
- D9.4** High-Level Architecture & Request Flows
- D9.5** Deep Dives, Scaling & Failure Handling
- D9.6** Trade-offs & Senior-Level Communication
- D9.7** Mocks, Deliberate Practice & Company Calibration

D10 — System Design Practice

30 NUMBERED SUBTOPICS

- D10.1** Design a URL Shortener
- D10.2** Design Pastebin
- D10.3** Design a Distributed Rate Limiter
- D10.4** Design a Key-Value Store
- D10.5** Design a Unique ID Generator
- D10.6** Design a Notification System
- D10.7** Design a News Feed
- D10.8** Design a Chat System
- D10.9** Design Google Drive / Dropbox
- D10.10** Design YouTube / Video Streaming
- D10.11** Design Search Autocomplete
- D10.12** Design a Ticket Booking System
- D10.13** Design Uber / Ride Hailing
- D10.14** Design a Payment System
- D10.15** Design a Distributed Job Scheduler
- D10.16** Design a Monitoring & Alerting Platform
- D10.17** Design a Distributed Cache
- D10.18** Design an Ad-Click Aggregation System
- D10.19** Design a Web Crawler
- D10.20** Design a Proximity / Nearby Service
- D10.21** Design Google Maps
- D10.22** Design a Distributed Message Queue
- D10.23** Design a Distributed Email Service
- D10.24** Design S3-Like Object Storage
- D10.25** Design a Real-Time Leaderboard
- D10.26** Design a Digital Wallet
- D10.27** Design a Stock Exchange
- D10.28** Design an Online Coding Judge
- D10.29** Design Collaborative Documents
- D10.30** Design an E-Commerce / Food Delivery Platform

Applied AI Engineering

LLM apps, RAG, evaluation — secondary until core is strong. 12 major groups · 77 sections.

E1 — Foundations & AI Tool Mastery

4 NUMBERED SUBTOPICS

- E1.1 Applied AI Landscape & Engineering Roles
- E1.2 AI Power-User & Coding-Agent Workflows
- E1.3 Python Essentials for AI Engineers
- E1.4 Python AI Backend & Data Stack

E2 — ML, Deep Learning & LLM Internals

8 NUMBERED SUBTOPICS

- E2.1 Machine Learning Foundations
- E2.2 Learning Paradigms & Core ML Problems
- E2.3 Training, Generalization & Classical Evaluation
- E2.4 Math for Applied AI — Just in Time
- E2.5 Classical ML with scikit-learn
- E2.6 Neural Networks & PyTorch Fundamentals
- E2.7 Transformers & Attention Internals
- E2.8 LLM Inference, Training Lifecycle & Limitations

E3 — LLM Application Engineering

5 NUMBERED SUBTOPICS

- E3.1 Provider APIs & Direct SDK Integration
- E3.2 Messages, Conversations & Streaming
- E3.3 Structured Outputs & Typed Extraction
- E3.4 Tool & Function Calling
- E3.5 Model Selection, Routing & API Reliability

E4 — Prompt & Context Engineering

5 NUMBERED SUBTOPICS

- E4.1 Prompt Fundamentals & Instruction Design
- E4.2 Context Engineering
- E4.3 Long Conversations & Memory Boundaries
- E4.4 Prompt Operations & Versioning
- E4.5 Output Validation & Prompt-Injection Awareness

E5 — Embeddings & Semantic Search

4 NUMBERED SUBTOPICS

- E5.1 Embedding Models & Representation Spaces
- E5.2 Similarity & Nearest-Neighbor Search
- E5.3 Vector Indexes & Storage
- E5.4 Semantic, Keyword & Hybrid Search

E6 — Retrieval-Augmented Generation

8 NUMBERED SUBTOPICS

- E6.1 RAG Architecture & Knowledge Ingestion
- E6.2 Chunking & Document Structure
- E6.3 Embedding & Indexing Pipelines
- E6.4 Retrieval & Reranking
- E6.5 Query Understanding & Retrieval Planning
- E6.6 Context Construction, Grounding & Citations
- E6.7 Advanced, Agentic & Multimodal RAG
- E6.8 RAG Evaluation & Failure Modes

E7 — Agents, Workflows & MCP

10 NUMBERED SUBTOPICS

- E7.1 Agents vs Deterministic Workflows
- E7.2 Planning, Decomposition & Verification
- E7.3 Agent Workflow Patterns
- E7.4 Agent State, Memory & Context
- E7.5 Agent Tool Use, Permissions & Approvals
- E7.6 Reliable Long-Running Agents
- E7.7 Multi-Agent Systems
- E7.8 MCP Architecture & Primitives
- E7.9 Building MCP Servers & Clients
- E7.10 Production MCP Security & Operations

E8 — Advanced Models & Multimodal AI

8 NUMBERED SUBTOPICS

- E8.1 Agent & RAG Frameworks—After Fundamentals
- E8.2 Hugging Face & Open-Weight Ecosystem
- E8.3 Local Models, Formats & Quantization
- E8.4 Model Serving & Inference Optimization
- E8.5 Fine-Tuning & Model Customization
- E8.6 Multimodal Document & Vision AI
- E8.7 Speech, Realtime & Voice Agents
- E8.8 AI Hardware & Distributed Inference

E9 — AI Evaluation & Testing

5 NUMBERED SUBTOPICS

- E9.1 Evaluation Strategy & Golden Datasets
- E9.2 LLM Output Evaluation
- E9.3 Retrieval & RAG Evaluation
- E9.4 Agent & Tool-Use Evaluation
- E9.5 Regression, Online Experiments & Feedback

E10 — Production AI & Product Engineering

7 NUMBERED SUBTOPICS

- E10.1 Production AI Reference Architecture
- E10.2 Streaming, Async Jobs & Backend Infrastructure
- E10.3 AI Performance, Caching & Context Efficiency
- E10.4 Reliability, Limits & Graceful Degradation
- E10.5 Model Routing & Cost Engineering
- E10.6 AI Observability & Quality Monitoring
- E10.7 AI Product Engineering & Decision Framework

E11 — AI Security & Responsible AI

5 NUMBERED SUBTOPICS

- E11.1 Prompt Injection & Untrusted Content
- E11.2 Agent Security, Agency & Tool Abuse
- E11.3 Data, Tenant & RAG Security
- E11.4 Input/Output Safety & Responsible AI
- E11.5 Adversarial Testing & AI Threat Modeling

E12 — Applied AI Project Ladder

8 NUMBERED SUBTOPICS

- E12.1 Project: Intelligent Document Analyzer
- E12.2 Project: Production Enterprise RAG
- E12.3 Project: AI Customer Support System
- E12.4 Project: Repository-Aware Coding Assistant
- E12.5 Project: AI Research Assistant
- E12.6 Project: Tool-Using Productivity Agent
- E12.7 Project: Recommendation & Personalization
- E12.8 Capstone: Multi-Tenant AI Executive Assistant